

Clones

Learning Objectives

- Understand what clones are and how they work in Scratch
- Create and delete clones dynamically during program execution
- Use clones to build multi-character game scenarios

Engage (5-7 minutes)

In farming, a farmer plants many identical seeds — each grows into the same type of plant. Clones in Scratch work similarly: one original sprite creates copies of itself, like a farmer sowing multiple seeds that all grow into the same crop, each clone identical to the parent.

Explore (10-12 minutes)

Students create a sprite and use “create clone” to make copies. They see each clone acts independently but shares the same costumes and scripts. Like identical twins in the village, clones look the same but can move and behave differently based on their own variables.

Explain (10-12 minutes)

Clones are temporary copies of a sprite that appear during program execution. When a clone is created, it receives the parent’s scripts, costumes, and sounds. Clones can be deleted when no longer needed. Like harvesting individual plants from a field, clones appear, perform their purpose, and can be removed.

Elaborate (8-10 minutes)

Students create a project where clicking creates clones that fall like raindrops. Each clone falls independently, creating a rain effect. Like monsoon droplets from clouds, clones multiply from one source and perform identical actions with slight variations.

Evaluate (5-8 minutes)

Can students create clones from an original sprite? Can they make clones behave independently? Do they understand when and why to use clones in a project?